

STM32F334

Digital Power Microcontrollers simplify design of high-efficiency power converters



Supports Complex Power Conversion topologies

- PFC Stage
 - Transition Mode PFC
 - Interleaved PFC
- DC-DC Stage
 - Half Bridge LLC Converters
 - Multiphase Interleaved LLC Converters
 - ZVS Full Bridge topology
- Secondary Side
 - Synchronous rectification for LLC topologies



500W Digital SMPS Available
2KW coming soon

Key Features:

ARM® Cortex-M4 at 72MHz with zero wait-state Routine Booster (CCM)

- Up to 90 DMIPS deterministic for real-time tasks

High-resolution timer with waveform builder and event handler (HRTIM)

- 217 ps high resolution (4.6 GHz equivalent) guaranteed on all channels vs voltage, temperature or manufacturing deviations
 - 10-channel timer made of 6 timings units that can be cross-coupled or work independently
- Advanced PWM waveform generation with minimized software
 - Smart functions, such as a hardware burst mode controller
- Complex event management
 - 10 external events inputs and 5 fault inputs
- Numerous interconnects

High-speed ADCs for precise and accurate control

- 12-bit SAR – 5 Msps, single-ended and differential inputs
- Sampling time down to 21 ns

Built-in analog peripherals for signal conditioning and protection

- Ultra-fast comparators (25 ns)
- 12-bit digital-to-analog converter (DAC)

Key Applications:

STM32F334 devices greatly simplify digital control of complex powersupply topologies used in several applications:



Authorized distributors

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